

# EEE4114F: Digital Signal Processing

## Class Test

14 March 2023

## SOLUTIONS

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Please do not put your name or student number anywhere on this script.

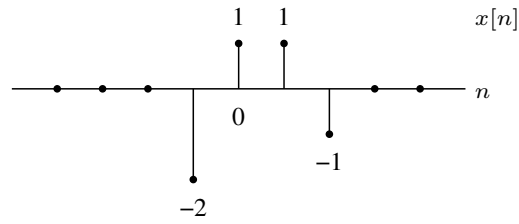
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### Information

- The test is closed-book.
  - This test has *four* questions, totalling 20 marks.
  - Answer *all* the questions.
  - You have 60 minutes.
  - An information sheet is attached.
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1. (5 marks) Consider the signal below:

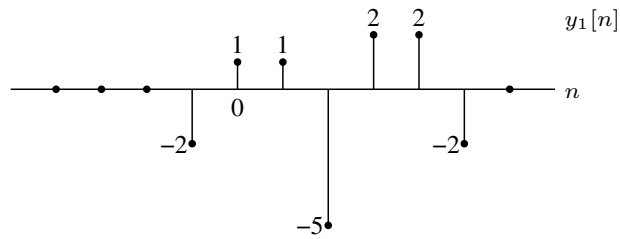


- (a) Sketch  $y_1[n] = \sum_{m=-1}^1 (m+1)x[n-3m]$ .  
 (b) Sketch  $y_2[n] = x[n] * x[-n]$ .

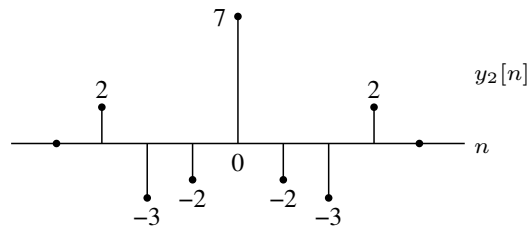
- (a) The expression can be written as

$$y_1[n] = \sum_{m=-1}^1 (m+1)x[n-3m] = 0 + x[n] + 2x[n-3],$$

and the plot is as follows:



- (b) Using graphical or other methods:



2. (5 marks) Consider a system with input  $x[n]$  and output  $y[n]$  that satisfies

$$y[n] = \begin{cases} x[n+1] & n > 0 \\ x[n-1] & n \leq 0. \end{cases}$$

- (a) Find the response of the system to a unit step  $x_1[n] = u[n]$ .
- (b) Find the response of the system to a shifted unit step  $x_2[n] = u[n-1]$ .
- (c) Is the system time invariant?
- (d) Prove that the system is linear.

(a) Observe that

$$y[-2] = x[-3], \quad y[-1] = x[-2], \quad y[0] = x[-1], \quad y[1] = x[2], \quad y[2] = x[3],$$

so the response to  $x_1[n] = u[n]$  is seen to be  $y_1[n] = u[n-1]$ .

- (b) The response to  $x_2[n] = u[n-1]$  can similarly be seen to be  $y_2[n] = u[n-1]$ .
- (c) The input  $x_2[n]$  is the input  $x_1[n]$  delayed by one sample, but we do not have  $y_2[n] = y_1[n-1]$ . Thus the system is not time invariant.
- (d) Suppose  $y_1[n]$  is the response to  $x_1[n]$  and  $y_2[n]$  is the response to  $x_2[n]$ . Then

$$y_1[n] = \begin{cases} x_1[n+1] & n > 0 \\ x_1[n-1] & n \leq 0 \end{cases} \quad \text{and} \quad y_2[n] = \begin{cases} x_2[n+1] & n > 0 \\ x_2[n-1] & n \leq 0. \end{cases}$$

Now, if  $y_3[n]$  the response to  $x_3[n]$  then

$$y_3[n] = \begin{cases} x_3[n+1] & n > 0 \\ x_3[n-1] & n \leq 0. \end{cases}$$

Consider the case where  $x_3[n] = ax_1[n] + bx_2[n]$ . Then

$$y_3[n] = \begin{cases} ax_1[n+1] + bx_2[n+1] & n > 0 \\ ax_1[n-1] + bx_2[n-1] & n \leq 0. \end{cases}$$

Now it is evident that  $y_3[n] = ay_1[n] + by_2[n]$ , so the system is linear.

3. (5 marks) Sketch the discrete-time signals with each of the following z-transforms:

(a)  $X_1(z) = (1 + 2z)(1 + 3z^{-1})(1 - z^{-1})$  with ROC all  $z$

(b)  $X_2(z) = \frac{(1+2z)(1+3z^{-1})}{(1-z^{-1})}$  with ROC  $|z| > 1$ .

(a) Multiplying out gives

$$\begin{aligned} X(z) &= (1 + 2z)(1 + 2z^{-1} - 3z^{-2}) \\ &= 1 + 2z^{-1} - 3z^{-2} + 2z + 4 - 6z^{-1} \\ &= 2z + 5 - 4z^{-1} - 3z^{-2} \end{aligned}$$

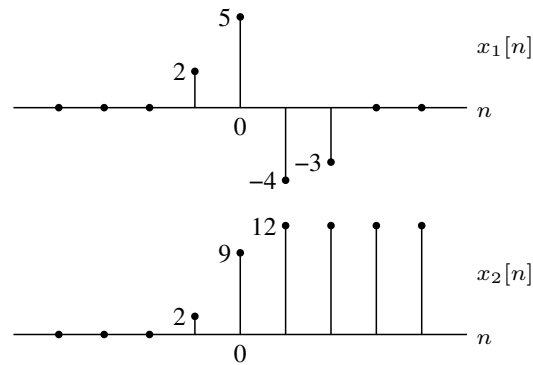
By inspection this inverts to  $x_1[n] = 2\delta[n+1] + 5\delta[n] - 4\delta[n-1] - 3\delta[n-2]$ .

(b) The given transform evaluates to

$$X_2(z) = \frac{2z + 7 + 3z^{-1}}{(1 - z^{-1})}$$

and the inverse is  $x_2[n] = 2u[n+1] + 7u[n] + 3u[n-1]$ .

Plots below:



4. (5 marks) We require a casual discrete-time LTI system with the property that if the input is

$$x[n] = (1/2)^n u[n] - 1/4(1/2)^{n-1} u[n-1]$$

then the output is

$$y[n] = (1/3)^n u[n].$$

Determine the impulse response  $h[n]$  and the system function  $H(z)$  of a system that satisfies these conditions.

The signals in the transform domain are

$$X(z) = \frac{1}{1 - 1/2z^{-1}} - \frac{1/4z^{-1}}{1 - 1/2z^{-1}} = \frac{1 - 1/4z^{-1}}{1 - 1/2z^{-1}}$$

with ROC  $|z| > 1/2$  and

$$Y(z) = \frac{1}{1 - 1/3z^{-1}}$$

with ROC  $|z| > 1/3$ . The system function is therefore

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} = \frac{1 - 1/2z^{-1}}{(1 - 1/3z^{-1})(1 - 1/4z^{-1})} = \frac{z(z - 1/2)}{(z - 1/3)(z - 1/4)} \\ &= \frac{a}{1 - 1/3z^{-1}} + \frac{b}{1 - 1/4z^{-1}} \end{aligned}$$

with  $a = -2$  and  $b = 3$ . The system has poles at  $z = 1/3$  and  $z = 1/4$ . For a causal system the ROC must extend outwards from the outermost pole, so the required ROC is  $|z| > 1/3$ . The impulse response is therefore

$$h[n] = -2(1/3)^n u[n] + 3(1/4)^n u[n].$$

We can (and should) check that the regions of convergence work out. To have  $Y(z) = H(z)X(z)$  we require that  $\text{ROC}_y$  contain  $\text{ROC}_h \cap \text{ROC}_x$ . Since  $\text{ROC}_h \cap \text{ROC}_x$  is  $|z| > 1/2$  we see that this is the case.

## Transforms

$$\begin{aligned}
 X(j\Omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt & \Longleftrightarrow & \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega & \text{(CTFT)} \\
 X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} & \Longleftrightarrow & \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega & \text{(DTFT)} \\
 X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} & \Longleftrightarrow & \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz & \text{(ZT)} \\
 X[k] &= \sum_{n=0}^{N-1} x[n]W_N^{kn} & \Longleftrightarrow & \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} & \text{(DFT)}
 \end{aligned}$$

## Discrete-time Fourier transform properties

| Sequences $x[n], y[n]$ | Transforms $X(e^{j\omega}), Y(e^{j\omega})$                                        | Property        |
|------------------------|------------------------------------------------------------------------------------|-----------------|
| $ax[n] + by[n]$        | $aX(e^{j\omega}) + bY(e^{j\omega})$                                                | Linearity       |
| $x[n - n_d]$           | $e^{-j\omega n_d} X(e^{j\omega})$                                                  | Time shift      |
| $e^{j\omega_0 n} x[n]$ | $X(e^{j(\omega - \omega_0)})$                                                      | Frequency shift |
| $x[-n]$                | $X(e^{-j\omega})$                                                                  | Time reversal   |
| $nx[n]$                | $j \frac{dX(e^{j\omega})}{d\omega}$                                                | Frequency diff. |
| $x[n] * y[n]$          | $X(e^{j\omega})Y(e^{j\omega})$                                                     | Convolution     |
| $x[n]y[n]$             | $\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$ | Modulation      |

## Common discrete-time Fourier transform pairs

| Sequence                                                                       | Fourier transform                                                                                        |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| $\delta[n]$                                                                    | 1                                                                                                        |
| $\delta[n - n_0]$                                                              | $e^{-j\omega n_0}$                                                                                       |
| 1 $(-\infty < n < \infty)$                                                     | $\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$                                                  |
| $a^n u[n] \quad ( a  < 1)$                                                     | $\frac{1}{1 - ae^{-j\omega}}$                                                                            |
| $u[n]$                                                                         | $\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$                      |
| $(n + 1)a^n u[n] \quad ( a  < 1)$                                              | $\frac{1}{(1 - ae^{-j\omega})^2}$                                                                        |
| $\frac{\sin(\omega_c n)}{\pi n}$                                               | $X(e^{j\omega}) = \begin{cases} 1 &  \omega  < \omega_c \\ 0 & \omega_c <  \omega  \leq \pi \end{cases}$ |
| $x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$ | $\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$                                            |
| $e^{j\omega_0 n}$                                                              | $\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$                                       |

### Z-transform properties

| Sequences $x[n], y[n]$ | Transforms $X(z), Y(z)$ | ROC                         | Property        |
|------------------------|-------------------------|-----------------------------|-----------------|
| $ax[n] + by[n]$        | $aX(z) + bY(z)$         | ROC contains $R_x \cap R_y$ | Linearity       |
| $x[n - n_d]$           | $z^{-n_d} X(z)$         | ROC = $R_x$                 | Time shift      |
| $z_0^n x[n]$           | $X(z/z_0)$              | ROC = $ z_0  R_x$           | Frequency scale |
| $x^*[-n]$              | $X^*(1/z^*)$            | ROC = $\frac{1}{R_x}$       | Time reversal   |
| $nx[n]$                | $-z \frac{dX(z)}{dz}$   | ROC = $R_x$                 | Frequency diff. |
| $x[n] * y[n]$          | $X(z)Y(z)$              | ROC contains $R_x \cap R_y$ | Convolution     |
| $x^*[n]$               | $X^*(z^*)$              | ROC = $R_x$                 | Conjugation     |

### Common z-transform pairs

| Sequence                                                                     | Transform                                                            | ROC                          |
|------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------|
| $\delta[n]$                                                                  | 1                                                                    | All $z$                      |
| $u[n]$                                                                       | $\frac{1}{1-z^{-1}}$                                                 | $ z  > 1$                    |
| $-u[-n-1]$                                                                   | $\frac{1}{1-z^{-1}}$                                                 | $ z  < 1$                    |
| $\delta[n-m]$                                                                | $z^{-m}$                                                             | All $z$ except 0 or $\infty$ |
| $a^n u[n]$                                                                   | $\frac{1}{1-az^{-1}}$                                                | $ z  >  a $                  |
| $-a^n u[-n-1]$                                                               | $\frac{1}{1-az^{-1}}$                                                | $ z  <  a $                  |
| $na^n u[n]$                                                                  | $\frac{az^{-1}}{(1-az^{-1})^2}$                                      | $ z  >  a $                  |
| $-na^n u[-n-1]$                                                              | $\frac{az^{-1}}{(1-az^{-1})^2}$                                      | $ z  <  a $                  |
| $\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$ | $\frac{1-a^N z^{-N}}{1-az^{-1}}$                                     | $ z  > 0$                    |
| $\cos(\omega_0 n) u[n]$                                                      | $\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$      | $ z  > 1$                    |
| $r^n \cos(\omega_0 n) u[n]$                                                  | $\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$ | $ z  >  r $                  |

### Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$